DO NOW — RECALL

Scope, modularity and parameters

Name: _____ Class: _____ Date: _____

1. If procedural Deposit(balance,5) returns a value that the caller ignores, does balance change?

2. What is a local variable?

3. How does a value-returning function pass a result back?

4. Why might using ref unexpectedly make a method harder to debug?

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

No. The returned value is discarded.

2. Model answer

A variable declared inside a method or block, with restricted scope.

3. Model answer

Using return.

4. Model answer

The method can change a caller's variable as a side effect, even if there is no returned result.

Vocabulary — know these terms

scope — Where a name can be accessed.

local — Variable with block or method-level scope.

global/class-level — Variable accessible in a wider scope, subject to access rules.

by value — Pass a copy of the value.

by reference — Pass an alias of the caller's variable.

NEW LEARNING

Read, annotate and explain

Copying an integer

Assignment b=a copies the integer 5, not a reference to a storage location.

Passing by reference

Increase(ref a) gives parameter value an alias of a, so the caller a changes to 8.

Use modular methods

A short reusable method has a clear input effect, can be tested, and reduces repeated code.

Explain one step in your own words, using the correct technical terms.

I DO — WORKED EXAMPLE

Follow the worked method

Trace AddFour(x) by value when caller x starts at 6.

Predict first

The function parameter number receives a copy of 6.

Model and check

Inside the function number becomes 10; caller x remains 6.

Model and check

The parameter leaves scope at the end of the call.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Method AddTwo(int n) adds 2 to n. Caller x=7 is passed by value. What is x after the call?

Question 2

Write an OCR-style subroutine Swap with two by-reference integer parameters and explain why by-reference is required.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Compare `int result=Double(x);` with `Double(ref x);` as ways of updating a number. Give advantages of one method for a pure calculation.

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

Passing an integer by value in C# changes the caller's integer when the parameter changes.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.